Materials Science & Engineering A 802 (2021) 140639

Contents lists available at ScienceDirect

Materials Science & Engineering A

journal homepage: http://www.elsevier.com/locate/msea

On the development of pseudo-eutectic $AlCoCrFeNi_{2.1}$ high entropy alloy using Powder-bed Arc Additive Manufacturing (PAAM) process

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ARTICLE INFO

Keywords:
High entropy alloys
$AlCoCrFeNi_{2.1}$
Additive manufacturing
Microstructure
Mechanical properties

ABSTRACT

A new Powder-bed Arc Additive Manufacturing (PAAM) processing which includes on-line remelting of deposited material has been developed for the manufacturing of high entropy alloys (HEAs) based on an existing $AlCoCrFeNi_{2.1}$ pseudo-eutectic system. The remelting process is typically applied in the arc melting process to improve the homogeneity of prepared material. We investigated the microstructure and mechanical properties of produced $AlCoCrFeNi_{2.1}$ HEA after applying a remelting process (1, 3, and 6 times) on each deposited layer. The results show the formation of the pseudo-eutectic microstructure, which consists of relatively large columnar grains of the dominant FCC phase (~90 wt%) and fine dendritic grains of the minor BCC phase (~10 wt%). The applied layer-remelting process shows negligible effects on the phase fractions and their compositions, however, it significantly degraded the tensile strength and ductility of prepared alloys. Particularly, the ductility of the alloy reduced dramatically from about 27% after one time layer-remelting to only about 3% after 3 times layer-remelting. This is rationalised by the significant localisation of thermally induced plasticity caused by repeated remelting of deposited material. We also show that this thermally induced plasticity leads to an increased amount of local misorientation in both constitute phases, which suggests an increased amount of stored dislocations in the microstructure. Despite the potentially strain hardening due to this accumulation of the thermally induced plasticity, the appreciable growth and constrained dendritic morphology of BCC grains that developed after remelting play a prevailing role on the materials strength, which limit the interfacial strengthening of the eutectic microstructure and consequently result in the loss of the tensile strength. The obtained results will assist in the further development and microstructure optimisation of novel HEAs using powder-based additive manufacturing processes.

1. Introduction

High entropy alloys (HEAs) are compositionally complex alloys which generally contain at least five major elements with percentages varying from 5 at.% to 35 at.%, forming a single- or dual-phase solid-solution crystalline structure despite that the constitution elements often have different crystal structures. Except for few Hexagonal Closed-Packed (HCP) single-phase HEAs [1], most of the HEAs tend to form the simple Face-Centred Cubic (FCC), or Body-Centred Cubic (BCC) single-phase solid-solution or a dual-phase (FCC + BCC), rather than complex compounds (such as intermetallics) owing to the high configuration entropy of the system [2,3]. HEAs have gained incredible attention from the research communities and industries globally since the first report by Yeh et al. [4] in 2004 owing to their unique conceptual designs and promising good microstructural stability and properties under various environmental conditions [5–8]. For example, HEAs are shown to have many attractive properties such as high hardness [9,10], good oxidation [11] and wear resistance [12], as well as excellent mechanical strength at elevated temperatures [13], likely correlated with their microstructural characteristics and large lattice distortion.

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https://doi.org/10.1016/j.msea.2020.140639
Received 17 August 2020; Received in revised form 3 November 2020; Accepted 3 December 2020
Available online 5 December 2020
0921-5093/

Currently, HEAs are considered as potential candidates in a wide range of applications, such as in aerospace, automotive and energy industries.

In addition to the conventional vacuum arc melting and casting approaches [14,15], powder-based Additive Manufacturing (AM) techniques, which are capable of manufacturing components with complex geometries, have been suggested for fabrication of various HEA compositions. Some attempts have been made for the AM of HEAs utilising powder materials and laser/electron beam heat sources [16–18]. For example, Brif et al. [19] reported the manufacturing of FeCoCrNi HEA using Selective Laser Melting (SLM) and the obtained HEA showed higher strength compared with the as-cast counterpart which was attributed to the fine microstructure achieved by SLM. Besides the rapid solidification effect, as demonstrated previously the Intrinsic Heat Treatment (IHT) associated with the AM process has a significant effect on the microstructure development of the AM-produced alloys [20–22], and thus contributes to the mechanical properties of the manufactured materials. Sistla et al. [23] reported the direct laser deposition of Al_x-$FeCoCrNi_{2-x}$ ($x = 0.3$ and 1) HEAs and found macro-cracks developed as a result of the mismatch in the thermal expansion coefficients between the deposited HEA and the substrate used. Additionally, Fujieda et al. [24] produced a dual-phase (FCC + BCC) AlCoCrFeNi HEA by selective electron beam melting and the cracks were observed at the interfaces of FCC and BCC grains due to their mechanical incompatibility. A recent (2020) comprehensive review on the advances of the laser and electron beam AM of HEAs has been presented in Ref. [25].

In order to explore the AM processing to achieve crack-free, high-performance HEA materials, a new Powder-bed Arc Additive Manufacturing (PAAM) approach was developed for the manufacturing of a model pseudo-eutectic $AlCoCrFeNi_{2.1}$ HEA. This system is selected considering its good melt flowability and the advantage of reduced compositional segregation during processing [26]. In addition, the remelting process used in conventional arc melting and casting process has demonstrated to be beneficial for improving the homogeneity in the microstructure and composition of the produced materials [27,28]. However, it remains unexplored whether this successful practice in arc melting can be utilised in our new PAAM process. Considering both the technical and fundamental importance, the present study focused on the evaluation of the effects of the layer-remelting process on the mechanical properties and microstructure of the prepared HEAs using PAAM. It is found that the repeated remelting processing has negligible influences on the composition and phase constitution of the prepared HEAs, whereas it results in significant degradation of tensile properties of the materials. This significant loss of ductility was believed to be attributed to the localisation of intragranular local misorientation developed in the microstructure in response to the repeated remelting processing, and the tensile strength degradation was correlated with the dendritic microstructural changes by remelting.

2. Experimental procedures

2.1. Materials and samples

Fig. 1 shows the experimental setup used for the Powder-bed Arc Additive Manufacturing (PAAM) of $AlCoCrFeNi_{2.1}$ HEAs, an example of the resultant $AlCoCrFeNi_{2.1}$ material as well as the sample extraction for characterisation in this work. As shown in Fig. 1a, a commercial (BOC®) inverter power source and a matching tungsten welding torch equipped with a 2.4 mm diameter non-consumable electrode were applied for the deposition. A mild steel plate with a good weldability was used as the substrate to ensure the stable deposition of initial few layers. The high purity (99.99 wt%) elemental powders of Al, Co, Cr, Fe and Ni with spherical morphologies and similar sizes of approximately 180 μm diameters were physically mixed in a ball miller according to a nominal composition of $AlCoCrFeNi_{2.1}$ (Al16.4-Co16.4-Cr16.4-Fe16.4-Ni34.4, at. %). The mixed powders were then fed into a designed hollow thin plate (1 mm thickness) powder holder in a manual manner in this proof-of-concept experiment. The arc current of 120 A and arc length of 3.5 mm were tested to be suitable for facilitating the steady deposition of powder materials and minimising the spattering. The torch travelling speed was set at 50 mm/min which demonstrated to be suitable for providing sufficient heat input to ensure complete fusion of powders during deposition. Furthermore, the argon shielding gas was supplied during deposition at a flow rate of 9 L/min to minimise the oxidation during the PAAM processing while not blowing away the powders in the powder bed. Additionally, a trailing argon gas shielding was used to protect the deposited layer from oxidation during the cooling stage. Analogous to the manufacturing of HAEs with the arc melting approach applying multiple remelting processing to eliminate the compositional segregation [27,28], we incorporated repeated on-line remelting processing in the PAAM of three individual thin-wall-structure

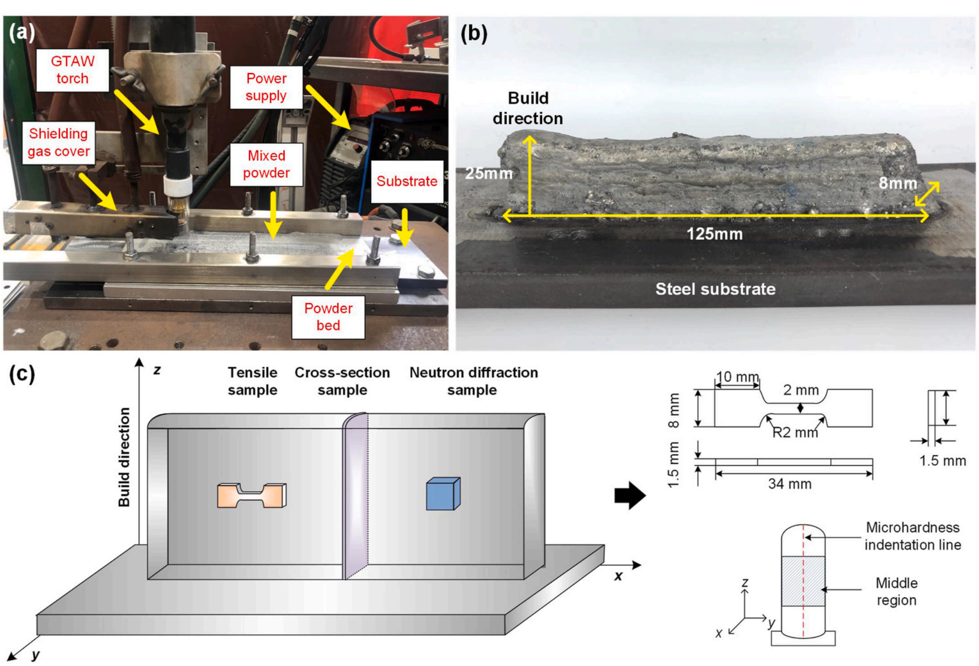


Fig. 1. (a) Experimental setup for the PAAM process used for the $AlCoCrFeNi_{2.1}$ high entropy alloy (HEA) preparation; (b) an example of the prepared thin-wall structured HEA material applying 6 times remelting process (HEA6); (c) schematics showing the extraction of samples for the tensile testing, metallographic observation and neutron diffraction measurement. The microhardness was measured along the build direction from the bottom to top regions of the cross-sectional (y-z plane) samples.

$AlCoCrFeNi_{2.1}$ HEA materials at once, 3 times and 6 times of remelting, respectively. Here on-line remelting means the remelting of each as-deposited layer during PAAM using the identical arc melting processing parameters without feeding a new layer of powders on top of the as-deposited layer. Three materials manufactured at different remelting times on each as-deposited layer are hereinafter referred to as HEA1 (once remelting), HEA3 (3 times remelting) and HEA6 (6 times remelting), respectively.

Fig. 1b shows a macrograph of the representative HEA6 thin-wall structured material manufactured using PAAM. The as-manufactured HEA6 sample of 15 deposited layers has a dimension of approximately $125 \times 25 \times 8$ mm^3 and typically shows no visible macro-cracking and distortion. The development of crack-free component in the present work is mainly attributed to the two aspects. Firstly, the interaction of powder materials with a much larger sized (3.5 mm length) arc beam in PAAM compared with the laser/electron beams in the selective laser/electron melting processes leads to the melting of a large amount of powders in a molten pool, facilitating their densification process. Also, the processing parameters particularly the travelling speed (as listed in Section 2.1) are optimised after a few trial experiments to avoid the discontinuous bead during processing. Secondly, the eutectic $AlCoCrFeNi_{2.1}$ composition showing reduced chemical segregation, compatible inter-phase boundaries (interfaces) and absence of detrimental phases [26,29] is selected, minimising the risk of cracking during manufacturing. Fig. 1c schematically shows the extraction of samples from the manufactured three HEA materials for mechanical, metallographic and neutron diffraction tests in order to investigate the effect of various remelting processing on the mechanical properties and microstructure of the prepared HEAs. Details of the sample preparation will be presented in the following sections (2.2 and 2.3).

2.2. Mechanical property testing

The tensile testing samples with the geometry and dimensions shown in Fig. 1c were extracted from the middle regions of the manufactured three HEA materials using electrical discharge machining (EDM). The tensile tests were performed at room temperature using an MTS 370 load frame at a constant crosshead displacement rate of 0.6 mm/min. In addition, as schematically shown in Fig. 1c, the Vickers microhardness measurements were carried out along a vertical indentation line from bottom to top regions (build direction) of the polished cross-sectional samples at an indentation interval of 0.5 mm with a load of 0.5 kg using a Struers DuraScan-70 automatic hardness tester.

2.3. Phase and microstructure analysis

In order to investigate the phase constitution in the bulk materials of the prepared three HEAs, neutron diffraction measurements were performed at room temperature using the high-intensity powder diffractometer Wombat [30] at the OPAL research reactor operated by the Australian Centre of Neutron Scattering (ACNS) of ANSTO. Neutron diffraction data were collected at a wavelength (λ) of 1.54 Å with a position sensitive detector during continuous oscillation ($\pm 2.5°$) of samples ($10 \times 10 \times 5$ mm^3) to improve the grain sampling statistics. Samples extracted from the representative middle regions (Fig. 1c) of three prepared HEA materials were used in the neutron measurements and each data set was collected over an exposure time of approximately 1 min. Quantitative phase analysis was performed using the Rietveld refinement program TOPAS Version 5 [31]. A spherical harmonics (up to 8^{th} order) model was applied in the refinement to account for the crystallographic texture in the manufactured HEA samples.

The y-z cross-sectional microstructure (Fig. 1c) was characterised using a range of microscopy techniques including Optical Microscopy (OM), Scanning Electron Microscopy (SEM) and Electron Backscatter Diffraction (EBSD) mapping. The preliminary OM observation was performed at the University of Wollongong to check the homogeneity of the microstructure along the build direction and to justify the representative region for further microstructural measurements in conjunction with the microhardness result. The Nikon Eclipse LV100NDA OM was used for this observation on the samples prepared using the standard metallographic techniques and procedures including mechanical grinding, polishing to 1 μm diamond suspension and followed by electrolytic etching at room temperature in a solution consisting of 10 ml oxalic acid and 100 ml distilled water at 6 V DC for 20 s. SEM observations were performed using the JEOL® JSM-6490LA SEM equipped with an X-Max Energy Dispersive X-ray Spectroscopy (EDS) detector operated at 20 kV accelerating voltage. The EDS chemical mapping was performed for 1 h over an area of approximately 120×90 μm^2 of the metallographic samples. Fracture morphologies of tensile test samples were also observed using the same SEM instrument. EBSD orientation mapping was performed at ANSTO using the Zeiss® UltraPlus™ SEM equipped with an Oxford Instruments® Aztec™ EBSD system and a Nordlys-S™ EBSD detector operated at 20 kV accelerating voltage. The EBSD orientation maps were collected on the HEA samples over an area of 800×800 μm^2 under 100x magnification using a scanning step size (h) of 1 μm, and over a smaller area of 100×100 μm^2 under 250x magnification at $h = 0.25$ μm. Analysis of the EBSD data was performed using the MTEX package [32]. Both the grain boundary misorientation and the intragranular local misorientation (LM) at the microstructural scale of the studied HEA samples were analysed using the EBSD data. The Kernel Average Misorientation (KAM) approach as described in detail by Kamaya [33] was used to compute the intragranular KAM (in degree) and construct the corresponding LM maps using the collected EBSD data. The KAM representing the average misorientation between the central pixel/orientation with all its equidistant most neighbouring pixels/orientations in the measured orientation map was determined. In this analysis only the neighbouring orientations belonging to the same grain and with a misorientation angle smaller than a threshold value of 7° were considered. Additionally, it has been previously shown that the KAM follows a lognormal distribution with its probability density function $f(\mathrm{KAM})$ expressed as [33,34]:

$$f(\mathrm{KAM}) = \frac{1}{(\pi)^{1/2}\sigma(\mathrm{KAM})}\exp\left\{\frac{-[\ln(\mathrm{KAM})-\mu]^2}{2\sigma^2}\right\} \tag{1}$$

where σ and μ are the shape and location parameters of the lognormal distribution, respectively. The mean (mKAM) and variance (νKAM) of the KAM distribution are defined as:

$$\mathrm{mKAM} = \exp\left(\mu + \frac{\sigma^2}{2}\right); \quad \nu\mathrm{KAM} = \exp\left(2\mu + \sigma^2\right)\left[\exp\left(\sigma^2\right) - 1\right] \tag{2}$$

The KAM distribution mean (mKAM) is then used to infer the level of the plastic strain at the microstructural scale of the studied HEA samples induced from the manufacturing process, and the distribution variance (νKAM) provides a measure of the strain localisation in the microstructure.

3. Results and discussion

3.1. Mechanical properties

As shown in Fig. 1c, the microhardness profile measurements were performed along the centreline of the cross-sectional HEA samples with diverse on-line remelting times. The results (not show here) reveal a relatively homogeneous microhardness distribution in the middle regions (2 mm–20 mm underneath the top deposition layer) of HEA1, HEA3 and HEA6 samples, with their average microhardness values of 245 ± 18 $HV_{0.5}$, 221 ± 10 $HV_{0.5}$ and 228 ± 13 $HV_{0.5}$, respectively. In contrast, the other regions (top and bottom along the build direction) show more significant variations in the microhardness. These results together with our OM inspections suggest that there are no significant variations in the microstructure and chemical composition of the middle regions of individual HEA samples investigated here. The microhardness variation in the top region (<2 mm below the top layer) is presumably due to the relatively less rapid thermal cycling and smaller temperature gradients during deposition of these last few layers and the resultant inhomogeneous microstructure. The dilution of the bottom region by the substrate, likely occurred during PAAM, is believed to be responsible for the observation of the microhardness fluctuation in this region. Similar observations have been reported in many previous studies on additively manufactured alloys including HEAs, e.g., on CrMnFeCoNi HEAs by SLM [35] and TiAl intermetallic by Wire-Arc Additive Manufacturing [36]. As the focus of this work is to understand the influence of on-line remelting processing on the mechanical and microstructural characteristics of the prepared HEAs, the relatively uniform middle region samples are used as representative samples for further tensile tests and microstructural observations.

Fig. 2a shows the representative engineering stress-strain curves and the measured Ultimate Tensile Strength (UTS), 0.2% offset Yield Strength ($\sigma_{0.2}$) and Elongation at fracture (ε) for the HEA1, HEA3 and HEA6 middle region samples. Fig. 2b presents the corresponding strain hardening rate curves. The results demonstrate that while the $\sigma_{0.2}$ of the studied three samples maintains at a similar magnitude (being 356 MPa for HEA1, 336 MPa for HEA3 and 388 MPa for HEA6), their UTS changes significantly with an increasing remelting time, showing a maximum of 719 MPa for HEA1 then a decrease to 414 MPa for HEA3 and subsequently a moderate increase to 508 MPa for HEA6. The comparable $\sigma_{0.2}$ of the three samples is consistent with their average microhardness result. Of particular note is that their Elongation at fracture (ε), which is a measure of the tensile ductility, decreases significantly after repeated remelting, from 27% for HEA1 to approximately 3% for both HEA3 and HEA6. Compared with the previous report on $AlCoCrFeNi_{2.1}$ HEA of UTS 1351 MPa and elongation 15% prepared by the conventional casting process [37], the HEA1 sample prepared by PAAM in this work shows an overall good comprehensive tensile properties with ε almost twice of that for the casting counterpart. The distinctive mechanical properties of HEAs achieved with the casting (an equilibrium process) and PAAM (non-equilibrium processing) techniques are attributed to the differences in the microstructures of materials under different processing. Notably, counter-intuitively against the observation of dramatically degraded ductility (indicative of actually lowered strain hardening capability) by remelting, very similar strain hardening rates are measured for HEA1 and HEA3, which are a bit lower than that for HEA6 (Fig. 2b). Detailed characterisation results on our HEA samples in terms of crystallographic phase, microstructure, chemical composition and fractography will be presented in the following sections to understand the remelting processing–microstructure–mechanical property correlation of the prepared HEAs by PAAM and to clarify the microstructural origin that leads to the significant changes of their tensile properties by repeated remelting processing.

3.2. Phase analysis

The $AlCoCrFeNi_{2.1}$ HEA has been previously shown to be an eutectic system with the FCC and BCC phase constitutions [26]. To confirm the phase constitution in the bulk of HEA1, HEA3 and HEA6 samples prepared by PAAM, neutron diffraction measurements were performed at room temperature and the obtained neutron diffraction patterns showing the measured reflections from the samples in a 2θ range of 40°–125° are presented in Fig. 3. All the three samples are found to comprise a dual-phase of FCC and BCC, confirming an eutectic microstructure of the investigated material which is consistent with previous reports on this system [26,38]. Due to the impact of preferred orientation (crystallographic texture) potentially developed during PAAM, the samples of various remelting times show different strongest reflections, e.g., a strongest {311} reflection is observed for HEA1 whereas {111} reflection is shown to be strongest for HEA3 (Fig. 3). The FCC ($Fm\overline{3}m$) and BCC ($Im\overline{3}m$) structural models were used in the Rietveld refinements of the collected neutron diffraction data. A typical Rietveld refinement fit of the HEA1 neutron data is shown in Fig. 3 and the refinement results of the three samples are summarised in Table 1. The refined lattice parameters of the FCC and BCC phases in the studied three samples are (i) HEA1: $a_{FCC} = 3.5877(1)$ Å, $a_{BCC} = 2.8690(3)$ Å; (ii) HEA3: $a_{FCC} = 3.5883(5)$ Å, $a_{BCC} = 2.8693(10)$ Å; and (iii) HEA6: $a_{FCC} = 3.5788(13)$ Å, $a_{BCC} = 2.8622(12)$ Å, showing slight variations due to repeated remelting processing. The lattice parameters obtained here are close to the corresponding values of as-cast $AlCoCrFeNi_{2.1}$ HEAs determined previously via X-ray diffraction [39,40]. Moreover, the phase quantification from the Rietveld refinement indicates that the FCC phase is a dominant phase in three samples with similar phase fractions of 87.9(5) wt.%, 89.3(5) wt.% and 89.5 (1.8) wt.% for HEA1, HEA3 and HEA6, respectively. This result appears to show that the repeated remelting processing only slightly alters the phase constitution of the present pseudo-eutectic HEAs, suggesting the formation of a relatively stable pseudo-eutectic microstructure via an isothermal pseudo-eutectic

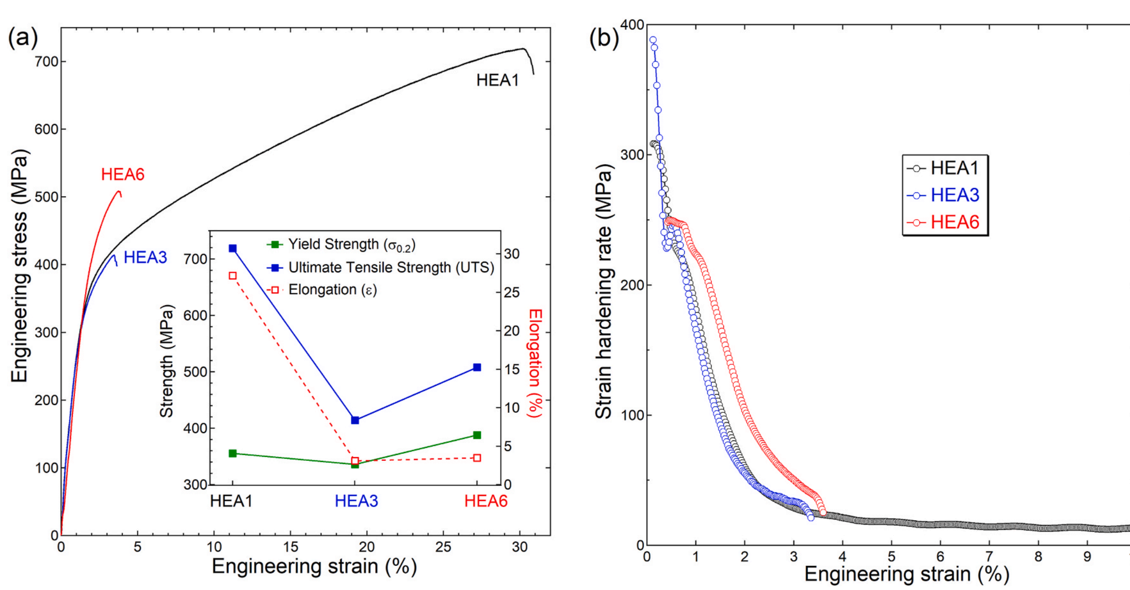


Fig. 2. (a) Engineering stress-strain curves together with the tensile strengths and elongation (inset) of HEA1, HEA3 and HEA6 middle region samples, and (b) the corresponding strain hardening rate curves.

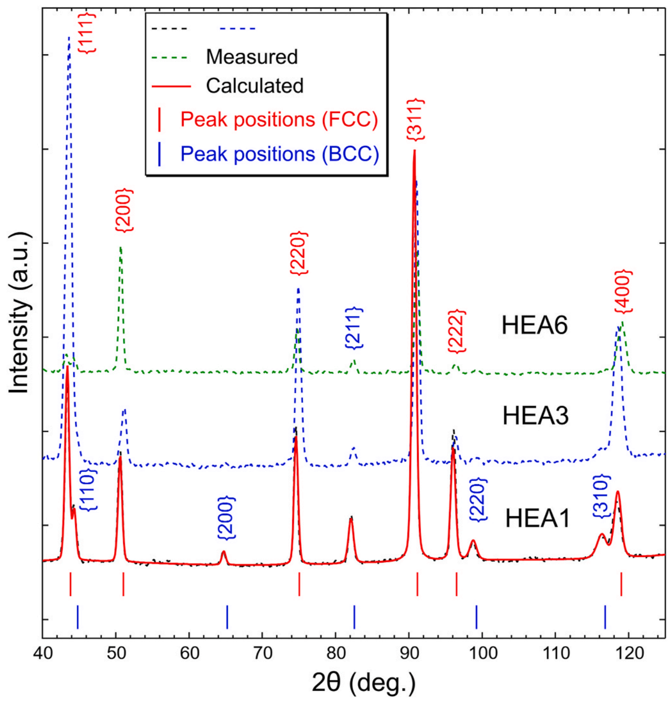


Fig. 3. Neutron diffraction patterns of the PAAM processed $AlCoCrFeNi_{2.1}$ HEA samples measured at room temperature and a representative Rietveld refinement fit of HEA1 data.

Table 1
Refined structural parameters of $AlCoCrFeNi_{2.1}$ HEA samples using neutron diffraction. The refined parameters are reported as: value (uncertainty in the last digit(s) of the value).

Samples		HEA1	HEA3	HEA6
a_{FCC}(Å)		3.5877(1)	3.5883(5)	3.5788(13)
a_{BCC}(Å)		2.8690(3)	2.8693(10)	2.8622(12)
Phase fraction (wt.%)	FCC ($Fm\bar{3}m$)	87.9(5)	89.3(5)	89.5 (1.8)
	BCC ($Im\bar{3}m$)	12.1(5)	10.7(5)	10.5 (1.8)
R_{wp} (%)		6.87	6.66	8.21
R_p (%)		5.36	4.28	5.93

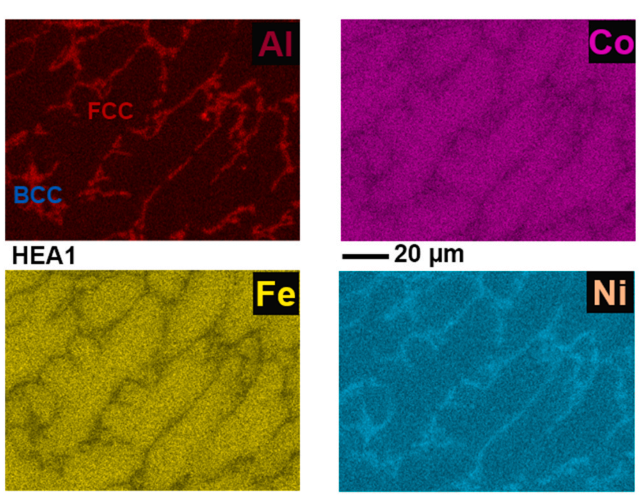
transformation (liquid phase L → FCC + BCC) at specific eutectic composition and temperature [26].

3.3. Microstructure and fractography

3.3.1. EDS mapping

EDS mapping was performed to further understand the microstructural characteristics of the prepared HEAs. Fig. 4 presents the representative Al, Co, Cr, Fe and Ni elemental EDS maps of the HEA1 middle region sample and a summary of the HEA1 compositional results by the EDS analysis. It is found that the dominant FCC and minor BCC phases contain all these 5 elements but show significant differences in their concentrations, allowing the phase constitution to be identified in the EDS maps. In conjunction with the quantitative EDS point scan results on each phase constitution, the BCC phase is observed to have much higher Al and Ni contents and to be less concentrated in Co, Cr and Fe elements compared with the FCC phase. Specifically, in the BCC phase Al and Ni contents are dominated, reaching up to 31.5 at.% and 40.2 at.%, respectively; the Cr and Fe have a comparable amount of ~10 at.% and Co is only about 6.8 at.%. In contrast, the FCC phase is rich in Ni (32.4 at.%) and Fe (21.9 at.%) and deficient in Al (11.1 at.%), and the rest Co and Cr have almost equal concentrations of ~17 at.%. In addition, the microstructural-averaged actual composition of the HEA1 sample measured from the EDS mapping is compared with the designed nominal composition. The results show that there is a loss of Al element of ~2 at.% while the measured contents of other elements coincide well with the designed values. This Al loss which has been observed in the Al-containing alloys prepared by arc melting [41] and casting process [42], is likely caused by the relatively high vapour pressure of Al [43], resulting in the evaporation during the PAAM process. It should be pointed out that the Al loss amount reported here (~2 at.%) is in experimental error due to the detection limit of the applied EDS analysis especially for light elements (≥1 at.% for Al). The measurements with other high sensitivity approaches such as X-ray fluorescence spectroscopy (XRF) mapping [44] may improve the accuracy of the compositional analysis.

Moreover, it is worth mentioning that the EDS mapping and elemental composition results are shown only for HEA1 in Fig. 4. Similar to the observation of slight changes in the FCC and BCC phase fractions after repeated remelting processing (Table 1), there are insignificant changes observable in the chemical compositions (<2 at.%) of HEA3 and HEA6 relative to HEA1. The stabilised microstructure observed in the present system in terms of the phase constitution and compositions is probably attributed to the fact that the pseudo-eutectic HEA system



Phase	Elements (at.%)				
	Al	Co	Cr	Fe	Ni
FCC	11.1	16.9	17.7	21.9	32.4
BCC	31.5	11.1	6.8	10.4	40.2
Average (measured)	14.3	16.7	16.9	17.1	35.1
Nominal	16.4	16.4	16.4	16.4	34.4

Fig. 4. Representative EDS maps showing the elemental distribution in the HEA1 sample, together with the compositional results of the FCC and BCC phase constitutions from EDS points scans and the microstructural-averaged from EDS mapping.

effectively reduces the compositional segregation during the manufacturing process [45]. Hence, it can be concluded that the remelting processing applied in this study does not cause any significant chemical composition change in the phase constitutions albeit that it did greatly alter the thermal histories of the samples and lead to significant degradation of their mechanical tensile properties.

3.3.2. EBSD microstructural analysis

Fig. 5 shows the EBSD phase maps, orientation maps as well as the grain boundary misorientation distribution histograms of the studied HEA1, HEA3 and HEA6 samples. The EBSD mappings were performed on the middle region samples along the build direction (Fig. 1c). It is clear from the EBSD maps that the microstructure of three samples is characterised by relatively large columnar grains (width ~10 μm and length ~10 to a few 100 s μm) of the dominant FCC phase and fine dendrite grains of the minor BCC phase. The fine BCC dendritic grains are distributed mostly in between the columnar FCC grains and show distinctive orientations from the adjacent FCC columnar grains. The observation shows a typical microstructure produced in HEAs by PAAM as observed in the additively manufactured conventional alloys [46,47]. The rapid solidification during the deposition of each layer leads to a significant temperature gradient within the molten pool, providing the driving force for the columnar and dendritic grain growth predominantly along the heat flow direction.

Moreover, the grain boundary misorientation distribution histograms shown in Fig. 5 indicate that there are predominantly the High Angle Grain Boundaries (HAGBs) with misorientation angles of >10° and a very low fraction of Low Angle Grain Boundaries (LAGBs, 2°–10°) existing in the microstructure of three samples. The HAGBs are observed at both the interphase (BCC – FCC) and intraphase (FCC – FCC and BCC – BCC) boundaries as we can see directly from the orientation maps as well as from the statistical analysis results of the grain boundary misorientation distributions in Fig. 5. Note that the grain boundary misorientation distribution is not sensitive to the repeated remelting processing, and similar results (Fig. 5) are observed for the investigated HEA1, HEA3 and HEA6 samples. In particular, it is noted that a majority of interphase (BCC – FCC) HAGBs at a misorientation angle of ~43° exist in the microstructure of all the three samples investigated. This characteristics resembles the microstructural feature commonly observed in ferrite-austenite dual-phase steels showing the well-known Kurdjumov-Sachs (K–S) orientation relationship $(\{111\}_{FCC\text{-}austenite}//\{110\}_{BCC\text{-}ferrite}$ and $<110>_{FCC\text{-}austenite}//<111>_{BCC\text{-}ferrite})$ characterised by an ideal misorientation of 42.85° between the two phases involved [48]. A supporting evidence for the compliance of this KS orientation relationship has been reported in the pseudo-eutectic $AlCoCrFeNi_{2.1}$ HEAs manufactured by arc melting process [40,49]. The EBSD misorientation distribution analysis result (Fig. 5) suggests that this KS orientation relationship is also present in the investigated $AlCoCrFeNi_{2.1}$ HEAs in the current work prepared using PAAM.

Note that additional thermally induced plasticity may occur in the

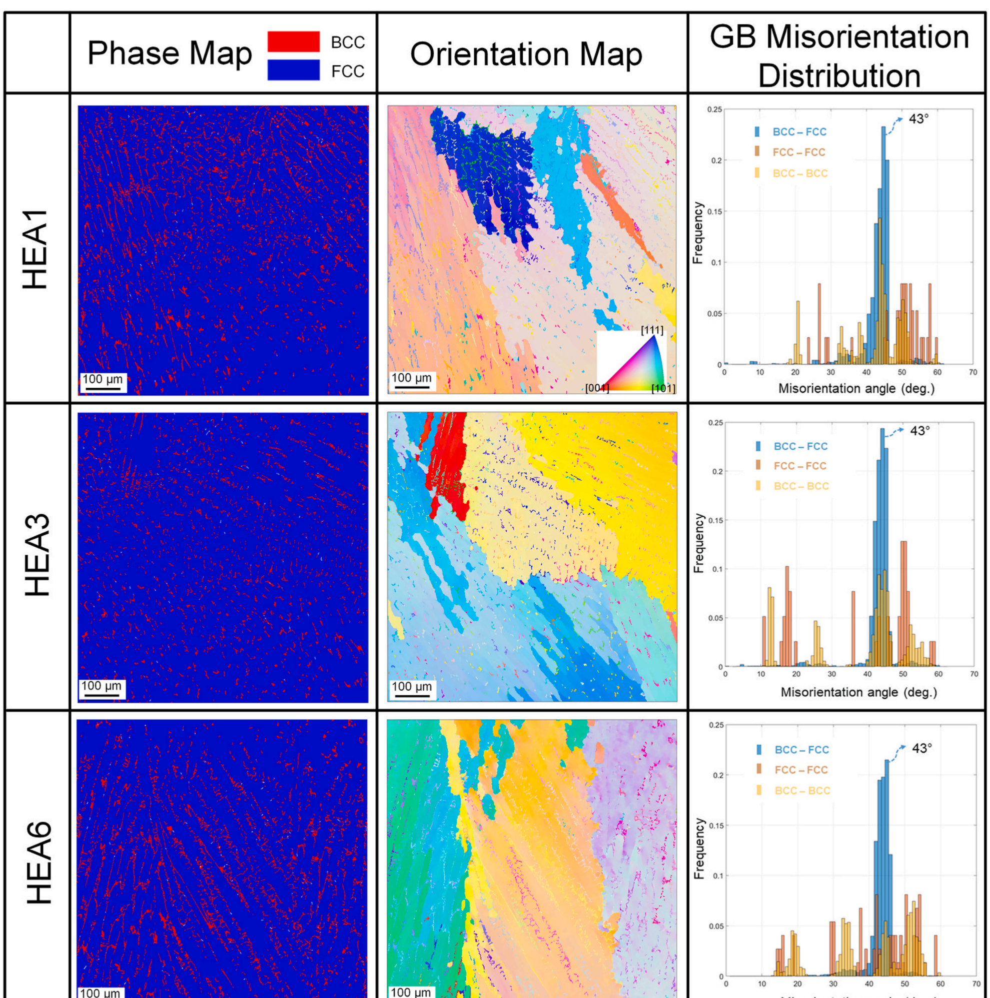


Fig. 5. EBSD phase maps, orientation maps and the measured grain boundary misorientation angle distributions of $AlCoCrFeNi_{2.1}$ samples including HEA1, HEA3 and HEA6.

prepared HEAs as a result of the applied repeated remelting process, besides that due to the inherent thermal cycling during the layer-by-layer deposition. The KAM local misorientation analysis is thus performed to evaluate the accumulation of plastic strain in the microstructure of the HEA1, HEA3 and HEA6 samples. The local misorientation is related to the amount of stored Geometrically-Necessary Dislocations (GNDs) in the microstructure. However, it needs to be emphasised that this EBSD-based analysis is insensitive to the presence of Statistically Stored Dislocations (SSDs), which have net-zero Burgers vector on the length-scale of the measurement. Fig. 6 then shows the obtained intragranular KAM maps together with the mean (mKAM) and variance (νKAM) of LM distribution in both present phase constituents – these were determined using the lognormal distributions (representative histograms of KAM distributions and lognormal fits for HEA1 are shown). It is clear from the KAM maps (Fig. 6a–c) that with repeated layer-remelting the intragranular KAM in the microstructure increases and the KAM distribution becomes more inhomogeneous. One can also observe that larger KAM appears to present in the fine BCC grains and in the vicinity of intraphase FCC – FCC grain boundaries (especially those boundaries of FCC grains with distinctive orientations, as shown in Figs. 5 and 6a-c). Furthermore, the determined mKAM and νKAM (Fig. 6d and e) provide quantitative measures showing statistically over the mapping areas that larger and more localised plastic strain builds up in both BCC and FCC grains with repeated remelting processing. Additionally, the BCC grains consistently show larger mKAM value than that of the FCC grains in three conditions, indicating more significant strain accumulation in BCC grains. To better visualise the LM along the boundaries and within the grains, higher magnification EBSD maps of three conditions showing the HAGBs and sub-grain boundaries in the FCC and BCC grains are presented in Fig. 7. Confirming the observations from the KAM maps in a lower magnification (Fig. 6), the distribution of these boundaries seen in Fig. 7 indicates that more localised LM tends to develop in the microstructure with repeated remelting. In addition, the morphology of BCC grains varies under different conditions. Both primary and secondary dendritic arms of BCC grains develop for HEA1 (Fig. 7a), while the primary dendritic BCC grains dominate for HEA3 (Fig. 7b) and the irregular BCC grains form for HEA6 (Fig. 7c). Compared with the fine BCC grains in the HEA1 sample, the slight (HEA3) to moderate (HEA6) growth of some BCC dendrites can be qualitatively identified. It is worth mentioning that the present observation with EBSD was able to capture relatively small local misorientation ($<1°$) as this technique is sensitive to a low amount of plastic strain [50]. Due to a relatively low amount of imparted plasticity and rather low resolution of neutron instruments, it was not possible to discern any strain broadening effect in the measured neutron diffraction patterns (Fig. 3).

The EBSD local misorientation KAM analysis (Fig. 6) implies that there is an increased amount of GNDs being stored in the microstructure of samples incorporating repeated remelting of deposited material. This is due to thermally induced plasticity generated. Despite this observation the degradation of the tensile strengths of the materials was observed after repeated remelting. The HEA1 sample without repeated remelting shows a maximum UTS of 719 MPa among three samples and its yield strength $\sigma_{0.2}$ (388 MPa) is comparable to that of HEA6 (356 MPa). In the eutectic $AlCoCrFeNi_{2.1}$, it has been shown that the FCC and BCC phase boundaries (interfaces) contribute significantly to the strength of the material [29]. We believe that the appreciable growth and constrained development of BCC dendrites as observed after repeated remelting (see Fig. 7) limit the interfacial strengthening and this effect alone results in the loss of the strength. This strength loss is presumably prevailing over the potential strain hardening induced by the accumulation of plastic strain (an increasing density of GNDs) during repeated remelting, and thus the overall tensile strengths of materials degraded after the remelting processing. Moreover, it is considered that

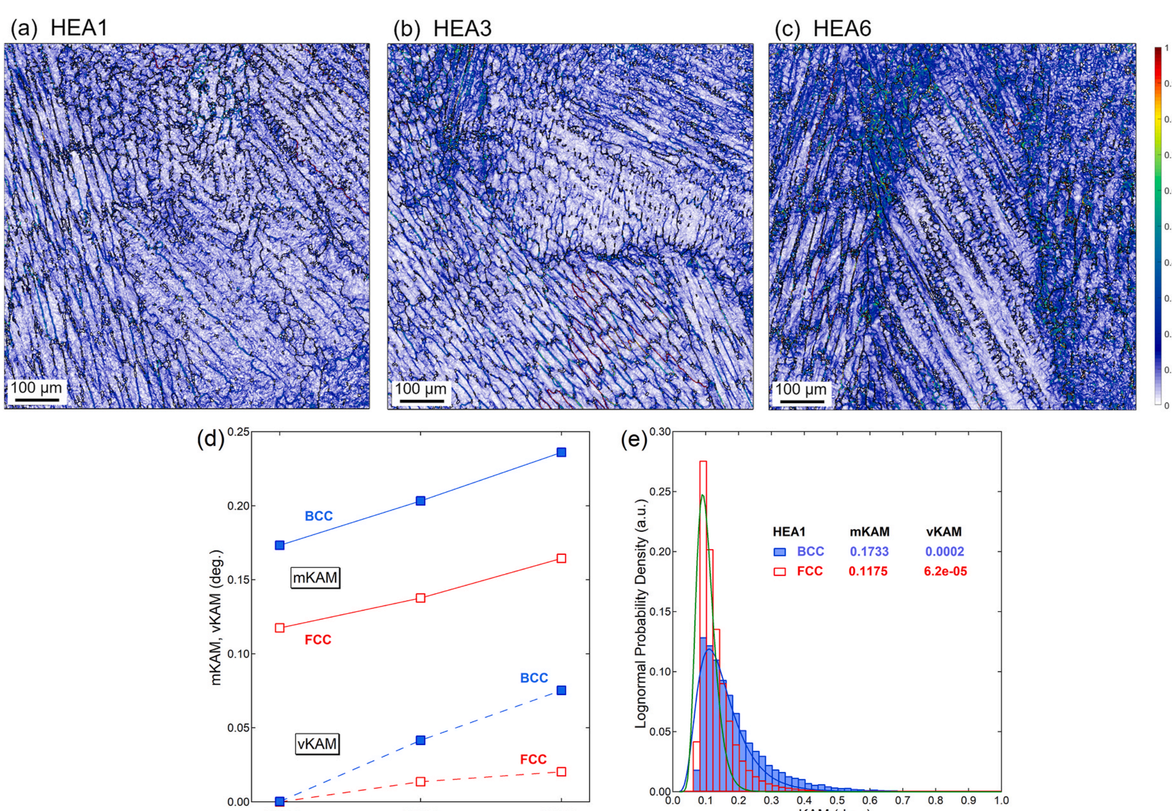


Fig. 6. Intragranular local misorientation (KAM) observed from the HEA1, HEA3 and HEA6 samples: (a–c) calculated KAM misorientation maps; and (d) mean (mKAM) and variance (νKAM) of KAM distribution in the FCC and BCC phases in the investigated three samples, obtained using their lognormal KAM distributions (a representative dataset of HEA1 is shown in (e)).

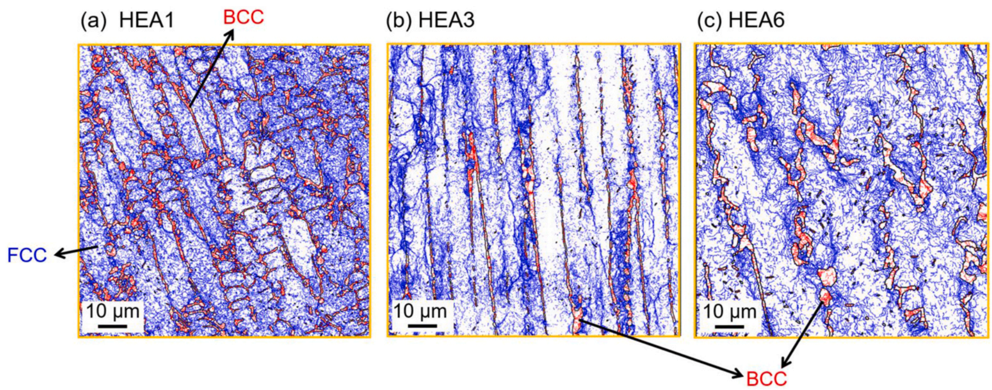


Fig. 7. High spatial resolution EBSD maps showing the grain boundaries in the microstructure of the (a) HEA1, (b) HEA3 and (c) HEA6 samples. The High Angle Grain Boundaries (HAGBs) (defined with the misorientation angle ≥10°) are shown in black while the sub-grain boundaries (low angle grain boundaries defined with the intragranular misorientation angle ≥0.1°) in FCC and BCC grains are coloured blue and red, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the localisation of the thermally induced plastic strain in the microstructure of HEAs after the repeated remelting (Figs. 6 and 7) induces discontinuity (micro-cracks) during tensile plastic deformation, limiting the capability of strain hardening and leading to the premature fracture. As a consequence, the tensile ductility of HEA3 and HEA6 both decreased significantly compared with the HEA1 sample. Apart from the thermally induced plastic strain, another consideration is that the oxidation is likely occurred during the out-of-chamber PAAM processing even with the inert gas shielding during deposition and after solidification, which may have potential effects on the mechanical properties of the deposited HEAs. Based on the evidences from previous investigations [51–53], the oxygen contamination during additive manufacturing of alloys is minimal and has negligible impact on the tensile strength and ductility of the deposited materials. Taking an example of Wire Arc Additive Manufacturing (WAAM) of titanium alloy Ti–6Al–4V (with strong affinity to oxygen), a negligible 70 ppm oxygen pick-up was measured during deposition using a localised trailing shield [52], which is much lower than the oxygen level of 1500 ppm in the original wire. Another study [53] reported that the oxygen content in the produced Ti–6Al–4V bulk alloy by WAAM with proper trailing shield configuration is similar to that of the starting wire while the poor trailing shield resulted in an oxygen pick-up of maximum 300 ppm during deposition. The statistical results of multiple tensile tests confirmed no significant difference in both tensile strength and ductility in the bulk Ti–6Al–4V alloys produced under different trailing shield conditions. Hence, it is reasonable to infer that the oxygen contamination potentially occurred during the present deposition processing with proper trailing shield is limited and we can rule out that it has any significance effect on the mechanical properties of the produced HEAs.

3.3.3. Fractographic observation

To understand the fracture micromechanisms of HEA1, HEA3 and HEA6 samples the fractographic observations were performed on the tensile tested samples and the results are shown in Fig. 8. The three samples appear to show similar intergranular cracking behaviour evidenced by the observation of the fracture traces (indicated by red arrowheads in Fig. 8). This crack propagation behaviour is linked to the columnar/dendritic microstructural characteristics with the LM localised near the grain boundaries and accumulated in the BCC grains, as observed for the HEA samples (Figs. 6 and 7). On the other hand, there are distinctive fracture features for the different samples. The shear slips and a large number of relatively deep dimples are observed on the fracture surface of the HEA1 sample (Fig. 8a1 and a2), indicating a typical ductile fracture consistent with the superior elongation observed in the tensile test (Fig. 2). The HEA3 sample shows both the cleavage facets and some dimples, demonstrating a mixed (brittle + ductile) fracture mode (Fig. 8b1 and b2). In contrast, the HEA6 displays a dominant brittle cleavage fracture morphology (Fig. 8c1 and c2). The transition of fracture mode (HEA1 ductile → HEA3 brittle + ductile → HEA6 brittle) from the fractographic observations further confirms the adverse effect of repeated remelting leading to the ductility degradation of $AlCoCrFeNi_{2.1}$ HEAs by PAAM.

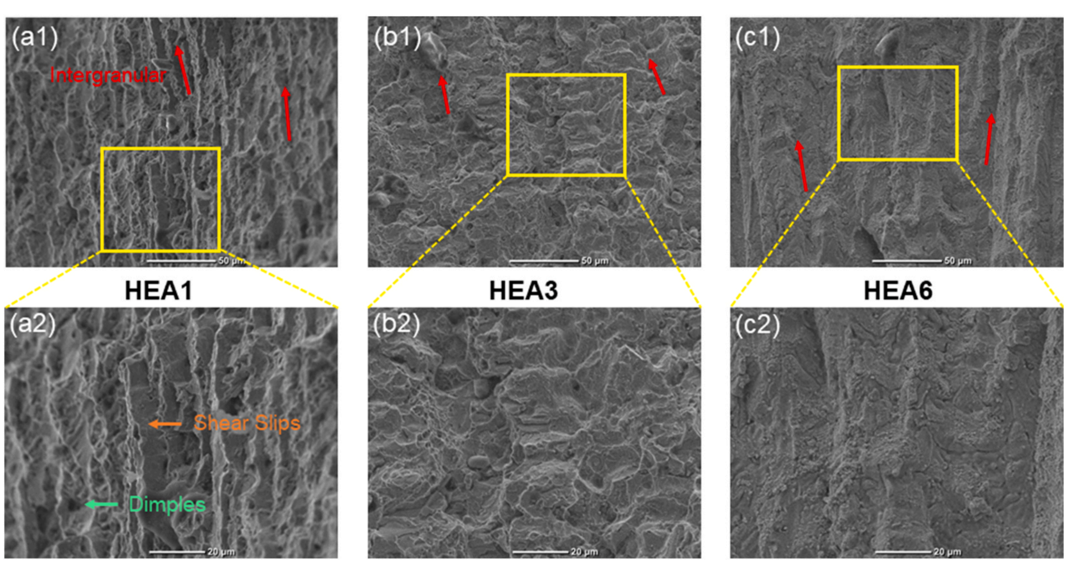


Fig. 8. SEM images showing the fracture surfaces of HEA samples after tensile testing: (a1, a2) HEA1; (b1, b2) HEA3 and (c1, c2) HEA6.

4. Conclusions

In the present work, the manufacturing of pseudo-eutectic $AlCoCrFeNi_{2.1}$ high entropy alloys (HEAs) using a novel Powder-bed Arc Additive Manufacturing (PAAM) approach, which incorporates on-line layer-remelting of deposited material has been investigated. The work focused on developing the understanding of the effect of the on-line layer-remelting on the microstructure and mechanical properties of produced samples. We prepared three alloy samples using developed PAAM process, where each deposited layer was remelted 1, 3 or 6 times (HAE1, HAE3, and HEA6). The main conclusions are summarised as follows:

(1) The representative $AlCoCrFeNi_{2.1}$ HEA material manufactured using PAAM with once remelting shows a good combination of tensile strength (UTS 719 MPa) and ductility (27%). This demonstrates the feasibility for the manufacturing of HEAs using proposed PAAM process.
(2) The additively manufactured $AlCoCrFeNi_{2.1}$ HEAs are confirmed to have a pseudo-eutectic microstructure characterised by relatively large columnar grains of the FCC phase (~90 wt%) and fine grains of the minor BCC phase (~10 wt%). The EBSD grain boundary misorientation analysis showed the Kurdjumov-Sachs (KS) orientation relationship between present FCC and BCC phases leading to about 43° misorientation.
(3) The applied on-line layer-remelting processing shows negligible influences on the phase fractions and compositions, however, it leads to a significant degradation of the mechanical tensile strength and ductility. In particular, the tensile ductility drops from about 27% in HAE1 (one time remelting of each deposited layer) to only 3% in HEA6 (6 times remelting of each deposited layer).
(4) We showed based on the EBSD analysis that repeated remelting of deposited material leads to the accumulation and localisation of thermally-induced plasticity across the microstructure as well as the appreciable growth of BCC grains and the development of constrained dendritic morphology of BCC grains. The latter effect is presumably prevailing on the material strength therefore resulting in the loss of the tensile strength.
(5) The localisation of thermally induced plasticity in the vicinity of grain boundaries is believed to be responsible for the observed degradation of ductility of the prepared alloy after application of 3 and 6 times layer-remelting.

Data availability statement

The raw/processed data required to reproduce these findings cannot be shared at this time as the data is used as part of an ongoing study.

CRediT authorship contribution statement

Bosheng Dong: Conceptualization, Investigation, Methodology, Validation, Writing - original draft. **Zhiyang Wang:** Funding acquisition, Investigation, Project administration, Supervision, Conceptualization, Methodology, Validation, Writing - original draft, Writing - review & editing. **Zengxi Pan:** Investigation, Supervision. **Ondrej Muránsky:** Investigation, Methodology, Writing - review & editing. **Chen Shen:** Funding acquisition, Investigation. **Mark Reid:** Investigation, Validation, Writing - review & editing. **Bintao Wu:** Investigation, Validation. **Xizhang Chen:** Investigation, Validation. **Huijun Li:** Funding acquisition, Investigation, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the Australian Nuclear Science and Technology Organisation (ANSTO) – University of Wollongong (UOW) Joint Project Seed Funding. The author B. Dong is supported by the China Scholarship Council (CSC) and C. Shen is supported by National Natural Science Foundation of China (NSFC, Funding No. 51901136). The authors acknowledge the Australian Centre of Neutron Scattering of ANSTO for provision of neutron beamtime under the experimental proposal 7044. The use of the electron microscopy facilities at both ANSTO and the Australian Institute for Innovative Materials (AIIM) of the University of Wollongong is also acknowledged. The technical support by Mr. Tim Palmer (from ANSTO) for the EBSD sample preparation is appreciated.

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